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Project 2 - Austin, TX Crime and Housing Data Analysis

Presentation Slides: [Link](#)

Project2 repository: https://github.com/loudpugpug42/cs5830_project2

Introduction

When looking for new housing, many have a commonly asked question, “Is this a safe neighborhood?” People obviously want to know how safe an area will be before committing to sign a lease or mortgage. In this project, we aim to understand more about the factors in an area that affect crime to help people make informed housing decisions, aid realtors in understanding these factors, and assist governments in their efforts to mitigate crime. Our sample was Austin, TX which serves as a good model for cities across the United States that are growing at a rapid pace with move-ins and new homes.

The data used provided information about criminal activity and housing in Austin, TX in 2015. We also utilized data with the population statistics of Austin to understand total numbers of people. To perform these analyses, we used a variety of standard statistical analyses such as computing averages and standard deviations, medians, and using Pearson R Coefficient Tests and T-Tests which were used to verify our findings as significant for stakeholders. Using these methods, we found that the characteristics of some areas such as high levels of unemployment, low housing affordability, and high poverty all lead to higher crime rates. We also determined there is a very strong correlation between the number of violent crimes and nonviolent crimes in a given area.

Dataset

The dataset contains crime records and housing data gathered in Austin, TX in 2015. Alongside the data is geographic information about each individual crime allowing it to be connected to its ZIP code. Useful attributes include the number of reported crimes, population below the poverty level, median household incomes and rent values, and additional measurements of housing affordability particularly for lower-income families and those working low-paying jobs in industries like retail and service. These attributes make the dataset suitable for analyzing the relationship between cost-of-living conditions and crime because they provide both a measure of criminal activity and indicators of economic hardship within the same ZIP codes. We also utilized data to know the total population of each ZIP code in the Austin area to perform some per-capita analysis.

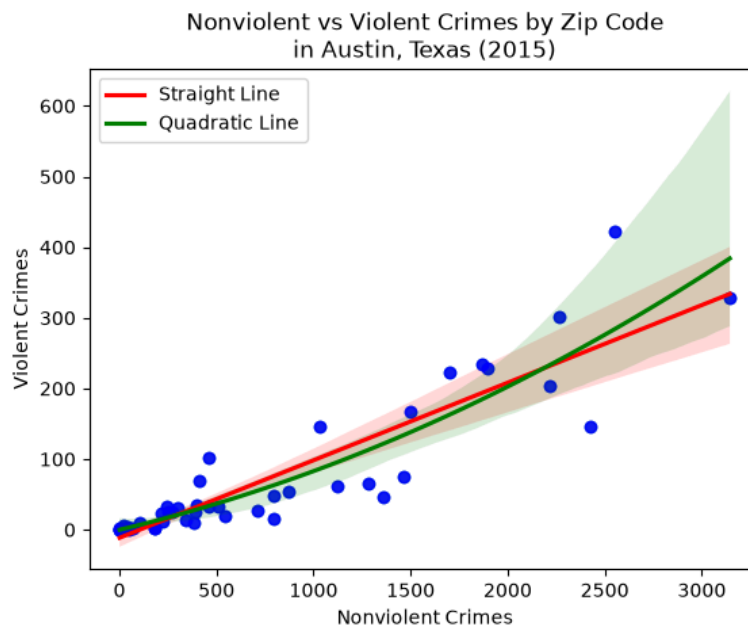
Analysis Technique

Various techniques were used to analyze the relationship between nonviolent and violent crimes in each ZIP code, including displaying scatterplots, calculating the correlation coefficient, and fitting regression lines to the data. We also used a bar chart to show the differences in total crimes per capita for areas with various levels of unemployment. Means and standard deviations, as well as Pearson R Correlation tests and T-tests, were utilized to help determine if employment status, poverty, or housing affordability within a ZIP code could be used to indicate

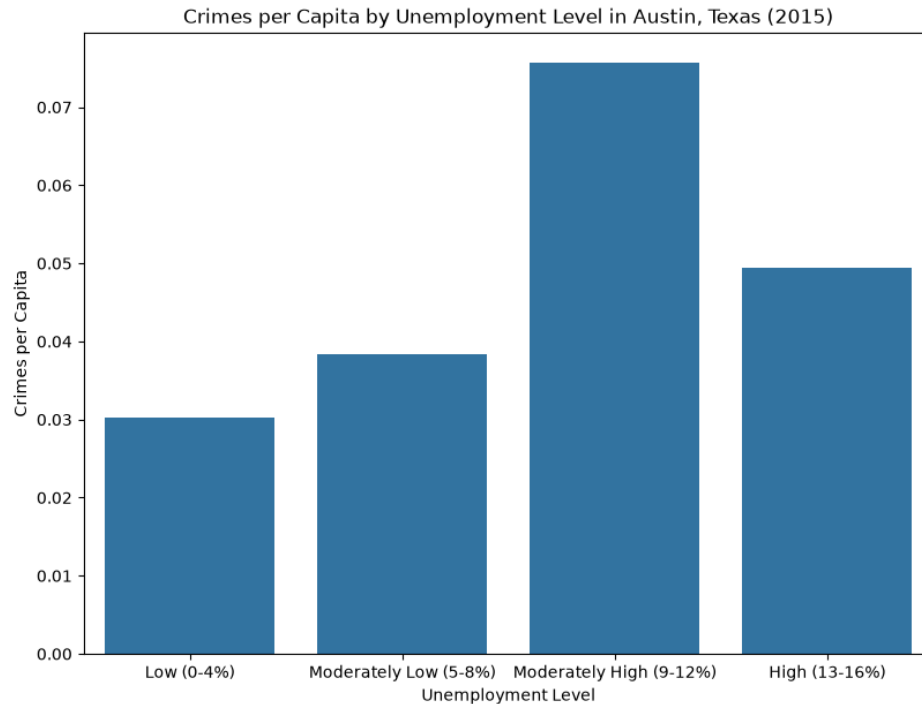
how prone each area was to criminal activity. These analyses were compatible with the data because both datasets we used contained ZIP code information and the geographic-based statistics were calculated at the ZIP code level.

Results

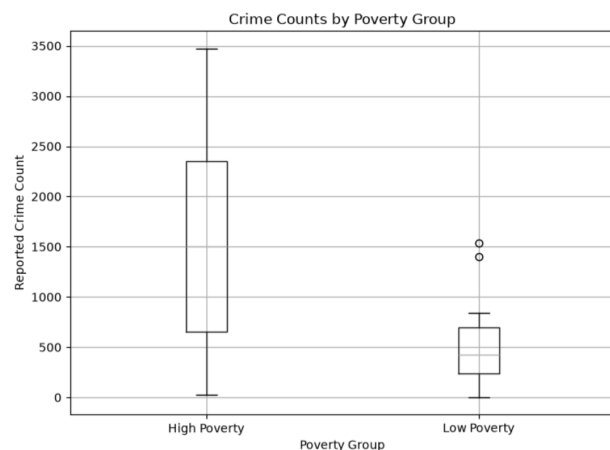
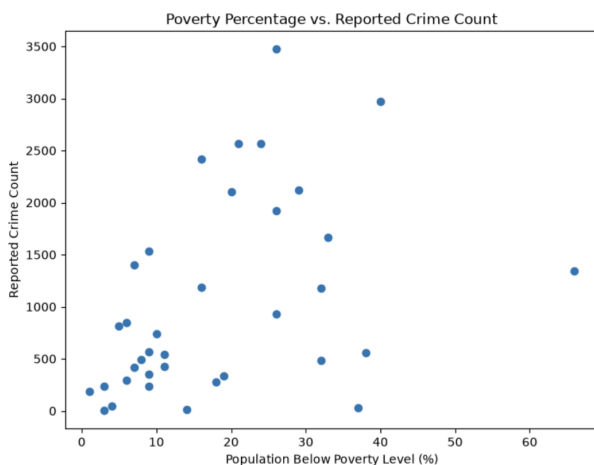
Our first analysis investigated the relationship between the numbers of violent and nonviolent crimes occurring in each ZIP code. We found that there was a very strong correlation between the two, resulting in a Pearson's R coefficient of 0.9 (with a p-value of 0). We also used both a straight and a quadratic regression line to describe the relationship and determined that the quadratic line fit the data slightly better than the straight line. This was determined using RMSE, which measures how far the data points tend to be from the line. The straight line had an RMSE of 42.2 and the quadratic line had an RMSE of 40.4 (smaller numbers indicate a better fit). The quadratic line shows that ZIP codes with a high crime rate tend to have a higher ratio of violent crime, which makes sense theoretically as well. This fact gives prospective homebuyers and renters more information about the safety of their possible neighborhood.



We also looked into how crime rates change in ZIP codes with differing levels of unemployment. As shown in the bar chart below, crimes per capita generally increase with the unemployment rate, but areas with a moderately high level of unemployment do have more crimes per capita than areas with high unemployment. This information can help individuals make housing decisions and help realtors accurately price homes based on the safety of the area.

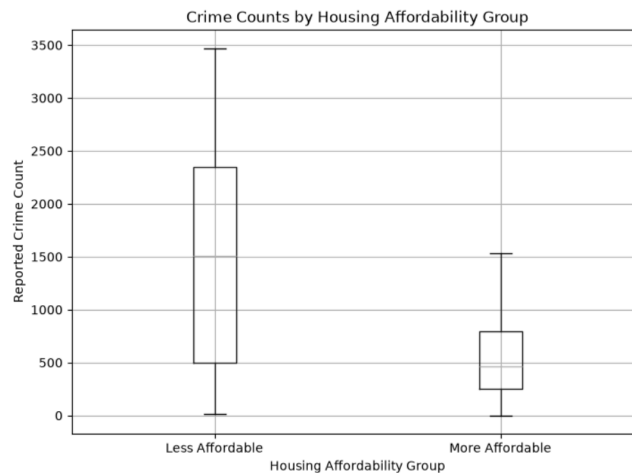


When analyzing the crime counts in areas with high populations below the median poverty level, two statistically significant results occurred. The first was indicates that ZIP codes with higher percentages of the population living beneath the poverty level share a correlation with the ZIP codes that have higher reported crime incidents. The other statistically significant result was that separating the codes by the median poverty level in Austin showed that ZIP codes within the lower percentiles clearly had more crime reported than those in the upper percentiles. Stakeholders can walk away with confidence in the result that higher poverty rates may indicate higher crime rates.



A similar analysis was done with housing affordability. Only one statistically significant result came, as there was no significant correlation found between the rent-to-income percentage of household income and the number of reported crimes. Outliers may affect this. Separating the

ZIP codes by the median rent-to-income percentage yielded a statistically significant result where ZIP codes with high rent housing compared to income had significantly more crime reported than housing with lower rent compared to income. Buyers in the region and those affected by these statistics should be confident in the notion that areas where more money is spent in rent are more likely to have higher crime counts. This makes sense because they who put too much income towards rent may resort to desperate measures to make ends meet.



	Analysis	Pearson_r	Pearson_p	T_statistic	T_test_p
0	Poverty vs. Crime	0.411416	0.012671	4.052122	0.000496
1	Housing Affordability vs. Crime	0.210775	0.217227	3.367902	0.002710

Technical

Originally, we thought that a major part of our analysis would be determining the best geographical unit to use to calculate aggregate crime statistics, but we quickly realized that all of the geography-based statistics were calculated using ZIP code. We then used ZIP code for all of the subsequent analyses, and were able to incorporate the ZIP code-level population statistics as well. The data needed some cleaning due to the large numbers of columns and inconsistencies with how some data was stored particularly in monetary columns. Some included dollar symbols (\$), some commas, some the percent symbol (%), but not all were the same. Specific definitions for violent or nonviolent crimes had to be looked up because the data did not make that distinction for us. The Pearson R Correlation and T-tests were good choices for our particular quantitative variables and separate distributions respectively. It led to one instance where something ruled statistically insignificant by the Pearson R Correlation test was considered statistically significant when the distributions were separated and tested by a T-test. The particular instance we refer to is the analysis of home affordability as an indicator for crime within a ZIP code, where separating the distribution by the median rent-to-income percentage exposed that zones where people who have a higher rent-to-income ratio (less affordable) have more crimes reported than zones with lower rent-to-income ratios.